Recovering the dynamics of a sequence seen only through noise

Results in brief — 24 September 2026

The question

A diffusion model learns by adding noise to data and learning to remove it. To remove it, it must learn one object at each noise level: the *score*. We study a case where the score can be computed exactly — data generated by a Markov chain, so the truth is available to compare against — and ask two things. What does the noise actually destroy? And is a method that knows the data has a chain structure better than a general-purpose network that does not, when both are shown exactly the same information?

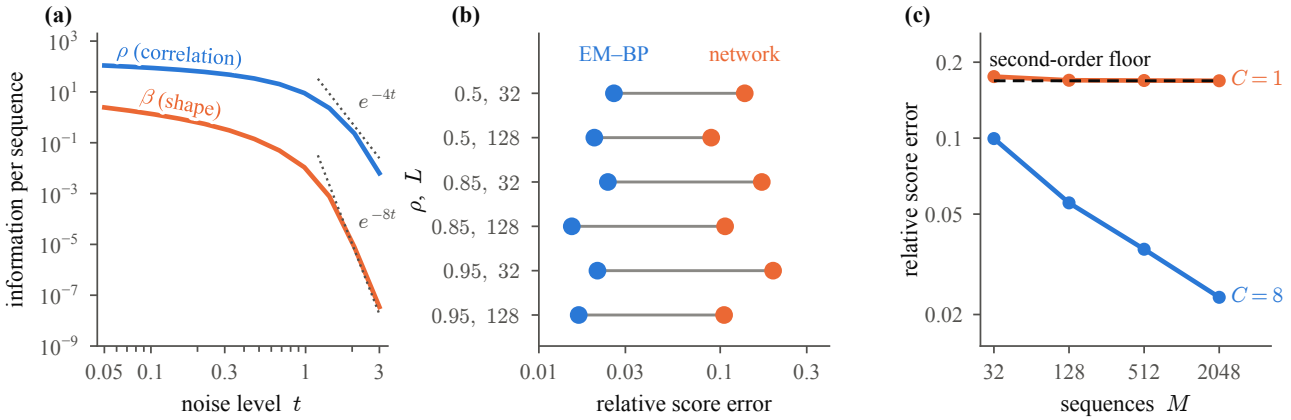


Figure 1: **(a)** What one noisy sequence still tells you about two properties of the chain as noise rises: its correlation ρ, and the *shape* of its randomness β (how non-Gaussian it is). Both fall exponentially but at different exponents; the dotted guides are the e^{-4t} and e^{-8t} the theory predicts. **(b)** Error of the structure-aware estimator against the best network at six combinations of ρ and sequence length L; both see one noisy view per sequence, never the clean one, and neither may use the true answer to decide when to stop. Lower is better. **(c)** Why structure helps: the dashed line is the best any second-order method can do, a chain model with one Gaussian innovation ($C = 1$) sits on it at every sample size, and one that can represent non-Gaussian structure ($C = 8$) goes far below.

What we found

Noise destroys fine structure far faster than it destroys correlation, at an exact rate. This is the theoretical result, and panel (a) is it measured. Each degree of structure decays at its own exponential rate, so the higher-order detail that makes data non-Gaussian goes first while correlation survives. Past a certain noise level everything still learnable is second-order — and we can say where that level is.

Knowing the structure is worth 5–10× at equal information. In panel (b) the chain estimator is ahead in all six regimes and in every one of the sixteen repetitions of each, by factors from 4.6 to 10.3. The comparison is deliberately unflattering to us: the networks are re-tuned separately in each regime, and our method may not peek at the true answer when deciding when to stop. The advantage is *smallest* where the chain carries the most non-Gaussian structure, which is the direction panel (a) predicts.

The advantage is fourth-order structure and nothing else. Panel (c) is the control. Restricted to Gaussian randomness, the chain model lands exactly on the second-order floor (0.168) and stays there however much data it gets; given the capacity for non-Gaussian structure it goes far below. The same holds in fifteen further configurations, the restricted model matching the theoretical floor to 0.0099 in the worst of them, and when the non-Gaussianity runs the *other* way, on a chain less heavy-tailed than Gaussian: the factor there is 8.8 [7.4, 10.2].

Where it stops working, and what it rests on

The estimator assumes a first-order chain, and we measured how wrong that may be before the advantage goes. Two departures reverse the ordering inside the range swept — a second order of memory, and a correlation varying along the sequence — at strengths 0.3–0.4 and 0.1–0.14; two others never reverse it. We report the boundary rather than bury it.

Every number here comes from a frozen experiment, regenerated from the raw outputs and independently recomputed by a second tool before this page was produced. One sample-complexity run is still on the cluster; nothing above depends on it. One correction is on the record: the capacity sweep in the earlier archival study is void — its training draw left every configuration fitted to a single sequence — and the capacity evidence above is independent of it.